

1) How many ways 10 boys and 10 girls can be arranged in a circular table so that boys and girls seat alternatively.

Sol) Given total no. of people = 20

Therefore, the no. of ways to arrange 10 boys and 10 girls around a

$$\text{circular table} = {}^{10}P_{10} \times {}^9P_9$$

$$= 10! \times 9!$$

2) Find n if i) $P(n, 2) = 72$

Sol) Given $P(n, 2) = 72$

$$n(n-1) = 72$$

$$n^2 - n - 72 = 0$$

$$n = 9, -8$$

n shouldn't be negative $\therefore n = 9$.

3) Find the no. of arrangements of letters in the word MAYAJALAM?

Sol) Given word MAYAJALAM

There are 2 M's, 4 A's, 1 Y, 1 J, 1 L Total = 9

$\therefore$ The total no. of arrangements of letters in the word MAYAJALAM

$$= \frac{9!}{2!4!1!1!1!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3}{2! \times 4!} = 7,560$$

4) State Pigeonhole Principle

Sol) Pigeonhole Principle: If there are more pigeons than pigeon holes then there must be at least one pigeon hole with at least 2 pigeons in it.

5) a) Find the expansion of $(x+y)^6$

Sol) Given $(x+y)^6$

$$\begin{aligned} \text{Expansion of } (x+y)^6 &= {}^6C_0 x^6 y^0 + {}^6C_1 x^5 y^1 + {}^6C_2 x^4 y^2 + {}^6C_3 x^3 y^3 + {}^6C_4 x^2 y^4 + {}^6C_5 x^1 y^5 + {}^6C_6 x^0 y^6 \\ &= x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6 \end{aligned}$$

b) How many ways even can 20 similar books be placed on 5 different shelves.

Sol) $r = 20, n = 5$

No. of ways can 20 similar books be placed on 5 different shelves

$$= \binom{n+r-1}{r} = {}^{20+5-1}C_{20}$$

$$= {}^{24}C_{20}$$

$$= {}^{24}C_4 = 10,626$$

c) How many different 8 digit numbers can be formed by arranging the digits 1,1,1,1,2,3,3,3?

sol) The no. of 8 digit numbers can be formed by arranging 1,1,1,1,2,3,3,3

$$= \frac{8!}{4!1!3!}$$

$$= \frac{8 \times 7 \times 6 \times 5}{3!}$$

$$= 280$$

c) How many integers between 10^5 and 10^6 ?

(a) Have no digits other than 2,5 or 8

(b) Have no digits other than 0,2,5, or 8

sol) There are 6 digit numbers b/w 10^5 & 10^6

$\overline{9} \quad \overline{10} \quad \overline{10} \quad \overline{10} \quad \overline{10} \quad \overline{10}$

There are 9×10^5 numbers b/w 10^5 & 10^6

a) There are 6 boxes

and each box can be filled with 3 ways

$\overline{3} \quad \overline{3} \quad \overline{3} \quad \overline{3} \quad \overline{3} \quad \overline{3}$

Total no. of digits other than 2,5 or 8 b/w 10^5 & $10^6 = 3^6$

b) There are 6 boxes

The first box can be filled with 3 ways.

$\overline{3} \quad \overline{4} \quad \overline{4} \quad \overline{4} \quad \overline{4} \quad \overline{4}$

and each of all other 5 boxes can be

filled with 4 ways.

Total no. of integers b/w 10^5 & 10^6 no digits other than 0,2,5 or 8

$$= 3 \times 4^5$$

7) In a class of 50 students 20 students play football and 16 students play hockey. It is found that 10 students play both the games. Find the number of students who play neither

sol) Let A be set of students playing football.

B be set of students playing hockey.

The $A \cap B$ represents set of students playing both games.

Given $|A| = 20$, $|B| = 16$, $|A \cap B| = 10$

By the principle of inclusion & exclusion $|A \cup B| = |A| + |B| - |A \cap B|$

$$= 20 + 16 - 10$$

$$= 26$$

Total strength = 50

No. of students who played neither = $50 - 26$

$$= 24$$

8) Prove that if any 30 people are selected, then we may choose a subset of 5 so that all 5 were born on the same day of the week?

sol) Given that no. of people (N) = 30

W.K.T no. of days per a week (K) = 7

By generalized pigeonhole principle $= \lceil \frac{N}{K} \rceil = \lceil \frac{30}{7} \rceil = 5$

Therefore we can choose a subset of 5 people all five were born on same day of the week.

9) a) How many different licence plates are there that involve 1 or 2 or 3 letters followed by 4 digits.

b) How many bit strings of length 8 contain (i) exactly five 1's?

(ii) an equal no. of 0's and 1's (iii) at least four 1's (iv) at least three 1's and at least three 0's

sol) a) The no. of licence plates are there that

involve 1 letter followed by 4 digits $= 26 \times 10^4$

The no. of licence plates are those that involve 2 letters followed

by 4 digits $= 26 \times 26 \times 10^4$

The no. of licence plates followed by 3 alphabets with

4 digits $= 26 \times 26 \times 26 \times 10^4$

Total ways $= 26 \times 10^4 + 26^2 \times 10^4 + 26^3 \times 10^4$

$= 702 \times 26 \times 10^4$

b) no. of bit strings of length 8 can

contain $= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$
 $= 256$

(i) The bit string of length 8 has 8

positions. These positions can be filled up with

exactly five 1's and three 0's in ${}^8C_5 \times 3C_3$ ways

$$= \frac{8 \times 7 \times 6 \times 5 \times 4}{1 \times 2 \times 3 \times 4 \times 5} = 56 \text{ ways}$$

(ii) the no. of ways for required bit strings

$= 8C_4 \times 4C_4$

$= 70 \text{ ways}$

(iii) the no. of ways for required bit strings

$$= {}^8C_4 \times 4C_4 + {}^8C_5 \times 3C_3 + {}^8C_6 \times 2C_2 + {}^8C_7 \times 1C_1 + {}^8C_8$$

$$= 70 + 56 + 28 + 8 + 1 = 163 \text{ ways.}$$

$1's$	$0's$
4	4
5	3
6	2
7	1
8	0

(iv) the bit string with at least 3 1's and there 0's

can be filled in following ways

three 1's five 0's

four 1's four 0's

five 1's three 0's

$$= {}^8C_3 \times 5C_5 + {}^8C_4 \times 4C_4 + {}^8C_5 \times 3C_3$$

$$= 56 + 70 + 56 = 182 \text{ ways}$$

$1's$	$0's$
3	5
4	4
5	3

10) consider the six digits 1, 2, 3, 5, 6 and 7. Assuming that repetitions are not permitted, answer the following: (i) How many four digit numbers can be formed from the six digits 1, 2, 3, 5, 6 and 7? (ii) How many of these numbers are less than 4000? (iii) How many of the numbers in (i) are even? (iv) How many numbers in (i) are odd? (v) How many of the numbers in (i) are multiple of 5?

b) Find the coefficient of $x^{10}y^9$ in $(2x-3y)^{200}$?

500) (i) The no. of 4 digit numbers can be formed from 6 digits 1, 2, 3, 5, 6, 7 is $= {}^6P_4 = 360$

(ii) A four digit number less than 4000

The first box can be filled in 3 ways only

(i.e., filled by 1 or 2 or 3)

second box can be filled in 5 ways

third box can be filled in 4 ways

fourth box can be filled in 3 ways

$\therefore$ The no. of 4 digit numbers which are less than 4000 $= 3 \times 5 \times 4 \times 3 = 180$

3	5	4	3
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(iii) 4 digit numbers which are even

last box can be filled in 2 ways

first box can be filled in 5 ways

second box can be filled in 4 ways

third box can be filled in 3 ways

$\therefore$ The no. of 4 digit numbers which are even $= 5 \times 4 \times 3 \times 2$

$$= 120$$

5	4	3	2
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(iv) 4 digit numbers which are odd
 last box can be filled in 4 ways.
 first box can be filled in 5 ways
 second box can be filled in 4 ways
 third box can be filled in 3 ways.

$\overline{5} \quad \overline{4} \quad \overline{3} \quad \overline{1, 3, 5, 7}$
 4 ways

$\therefore$ The no. of 4 digit numbers which are odd $= 5 \times 4 \times 3 \times 4 = 240$

(v) 4 digit numbers which are multiple of 5

last box in 1 way
 first box in 5 ways
 second box in 4 ways
 third box in 3 ways.

$\overline{5} \quad \overline{4} \quad \overline{3} \quad \overline{5}$
 1 way

$\therefore$ The no. of 4 digit numbers which are multiple of 5 $= 5 \times 4 \times 3 = 60$

(vi) coefficient of $x^{101}y^{99}$ in $(2x-3y)^{200}$

write General term (or) $(r+1)$ th term in $(x+y)^n = {}^nC_r x^{n-r} y^r$
 $= {}^{200}C_r (2x)^{200-r} (-3y)^r$
 $= {}^{200}C_r (2)^{200-r} (x)^{200-r} (-3)^r y^r$
 $= {}^{200}C_r (2)^{200-r} (-3)^r x^{200-r} y^r$

$x^{200-r} y^r = x^{101} y^{99} \Rightarrow x^{200-r} y^r = {}^{200}C_r 2^{200-r} (-3)^r$

coeff of $x^{101}y^{99} = {}^{200}C_{99} 2^{101} (-3)^{99} 2^{200-99} = 101, r=99$
 $= - \frac{200!}{(101)!(99)!} \cdot 2^{101} 3^{99}$

11) a) Suppose a department consists of eight men and nine women. In how many ways can we select a committee of (i) three men and four women? (ii) four persons that has at least one woman?
 b) How many integral solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6, x_5 \geq 0$

200) a) i) Out of 8 men, three men are selected $\Rightarrow {}^8C_3$
 out of 9 women 4 women are selected $\Rightarrow {}^9C_4$

$\therefore$ The no. of ways to select 3 men & 4 women
 $= {}^8C_3 \times {}^9C_4 = 7056$

(ii)

8M	9W	
3	1	1W $\Rightarrow$ 3M
2	2	2W $\Rightarrow$ 2M
1	3	3W $\Rightarrow$ 1M
0	4	4W $\Rightarrow$ 0M

The required ways to select atleast 1 woman
 $= {}^8C_3 \times {}^9C_1 + {}^8C_2 \times {}^9C_2 + {}^8C_1 \times {}^9C_3 + {}^8C_0 \times {}^9C_4$
 $= 2310$

$$11b) x_1 + x_2 + x_3 + x_4 + x_5 = 20$$

$$x_1 - 3 + x_2 - 2 + x_3 - 4 + x_4 - 6 + x_5 - 0 = 5$$

$$\text{where } x_1 - 3 \geq 0 \quad x_2 - 2 \geq 0 \quad x_3 - 4 \geq 0 \quad x_4 - 6 \geq 0 \quad x_5 \geq 0$$

$$x_1 - 3 = y_1, x_2 - 2 = y_2, x_3 - 4 = y_3, x_4 - 6 = y_4, x_5 = y_5$$

$$y_1 + y_2 + y_3 + y_4 + y_5 = 5 \quad \text{where } y_i \geq 0$$

$$= C(n+r-1, r) \text{ the } r=5, n=5$$

$$\Rightarrow C(5+5-1, 5) = C(9, 5) = 9C_5$$

$$\Rightarrow 9C_4 = 126.$$

12) a) Consider a set of integers from 1 to 250. Find how many of these numbers are not divisible by 3 or 5 or 7.

b) Find the coefficient of $u^2 w^3 x^4 y^2$ in the expansion of $(u+v+2w+x+3y+z)^{11}$

sol) a) Let A, B, C be set of positive integers from 1 to 250

$$|A| = \left\lfloor \frac{250}{3} \right\rfloor = 83$$

$$|B| = \left\lfloor \frac{250}{5} \right\rfloor = 50$$

$$|C| = \left\lfloor \frac{250}{7} \right\rfloor = 35$$

$$|A \cap B| = \left\lfloor \frac{250}{15} \right\rfloor = 16$$

$$|A \cap C| = \left\lfloor \frac{250}{(3)(7)} \right\rfloor = 11$$

$$|B \cap C| = \left\lfloor \frac{250}{(5)(7)} \right\rfloor = 7$$

$$|A \cap B \cap C| = \left\lfloor \frac{250}{3 \times 5 \times 7} \right\rfloor = 2$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

$$= 83 + 50 + 35 - 16 - 7 - 11 + 2$$

$$= 136$$

The no. of positive integers from 1 to 250 which are not divisible by 3, 5 or 7

$$= 250 - 136$$

$$= 114$$

$$b) (x_1 + x_2 + \dots + x_t)^n = \sum P(n; k_1, k_2, k_3, \dots, k_t) x_1^{k_1} x_2^{k_2} \dots x_t^{k_t}$$

$$\Rightarrow (u+v+2w+x+3y+z)^{11} = \sum P(11; 2, 3, 4, 2, 0) u^2 (2w)^3 x^4 (3y)^2 z^0$$

$$\text{coefficient of } u^2 w^3 x^4 y^2 = 72 \cdot \frac{11!}{2! 3! 4! 2!}$$

$$= 72 \cdot \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5}{2 \times 3 \times 2 \times 2}$$

$$= 4989600$$